



Lab No 1

Cloud Computing

**Lab Title: GitHub Identity, Student Portal Registration,
and Ubuntu Server Installation**

Submitted To: Engr.Waqas Saleem

Submitted By: Laiba Saleem

Registrartion No: 2024-BSE-066

Course Code: SE-665

Section: B

GitHub Username: LaibaSaleem066

**DEPARTMENT OF SOFTWARE ENGINEERING
SESSION (2024-2028)**

Task 1

Create and Configure Your GitHub Account

Description:

Set up a GitHub account under the username LaibaSaleem066 and updated the public display name to Laiba Saleem within the account settings.

The screenshot shows the GitHub profile page for user LaibaSaleem066. The header includes the username, a search bar, and navigation links for Overview, Repositories, Projects, Packages, and Stars. The profile section features a circular avatar with a teal and white geometric pattern, the name 'Laiba Saleem', the username 'LaibaSaleem066', and an 'Edit profile' button. Below the profile is a 'Popular repositories' section with the message 'You don't have any public repositories yet.' To the right is a '1 contribution in the last year' section with a calendar grid showing a single contribution in September. Below the calendar is a 'Contribution settings' dropdown and a 'Learn how we count contributions' link. At the bottom, there is a 'This is your contribution graph' section with a brief explanation of the graph and a link to 'additional contributions'.

Laiba Saleem
LaibaSaleem066
Edit profile

Popular repositories

You don't have any public repositories yet.

1 contribution in the last year

Contribution settings

Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug

Mon
Wed
Fri

Learn how we count contributions

Less More

This is your **contribution graph**. Your first square is for joining GitHub and you'll earn more as you make [additional contributions](#). More contributions means a higher contrast square color for that

Laiba Saleem (LaibaSaleem066)
Your personal account

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Public profile

Name

Laiba Saleem

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Public email

Select a verified email to display

You can manage verified email addresses in your [email settings](#)

Bio

Profile picture

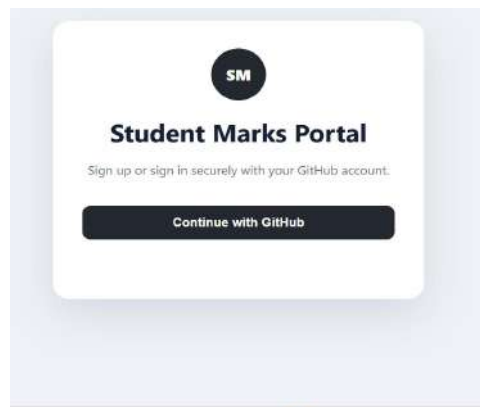
Edit

Task 2

Register in the Student Marks Portal

Description:

Logged into the Student Marks Portal using GitHub authentication and enrolled in Section B of Cloud Computing (SE-665) with Registration No 2024-BSE-66.



STUDENT MARKS PORTAL

Academic Record

Sign out

**LaibaSaleem066**
GitHub authenticated

Registration

2024-BSE-66

GitHub username

[LaibaSaleem066](#)

Email

fa24b1-se-066@fjwu.edu.pk

Approved courses

1

+ Enroll in another course

APPROVED COURSES

Course Marks

Published

Cloud Computing (SE-665) Section B

Class Participation

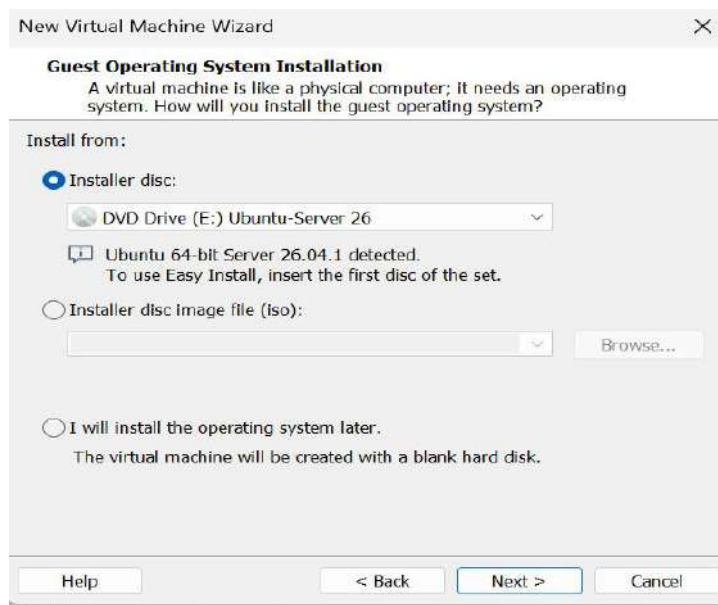
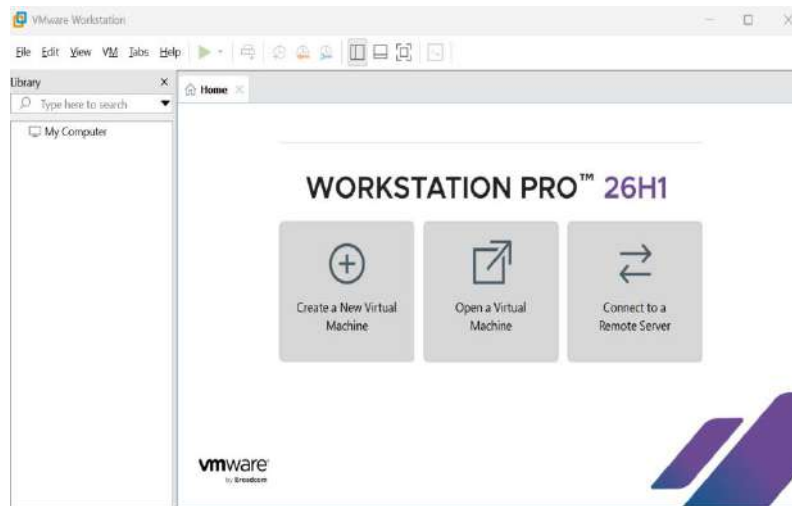
ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
Class Participation	0	10	0.0%
Total	0	10	0.0%

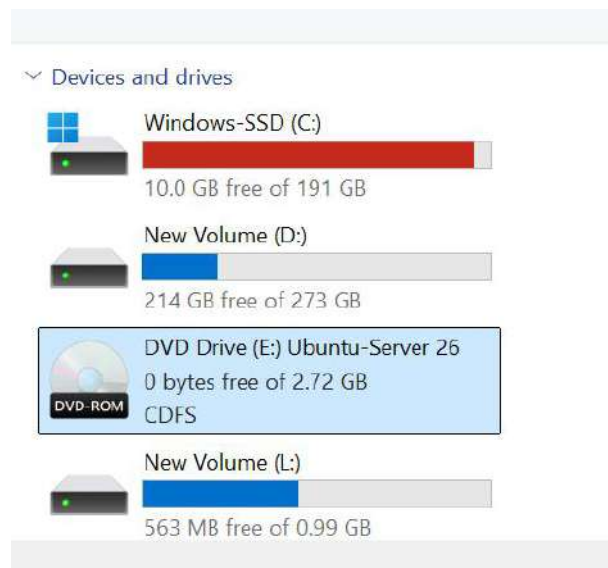
Task 3

Download VMware Workstation Pro and Ubuntu Server

Description:

Installed VMware Workstation Pro on the Windows host machine, downloaded the latest Ubuntu Server LTS ISO image, and verified that at least 20 GB of free storage space was available for the virtual machine.



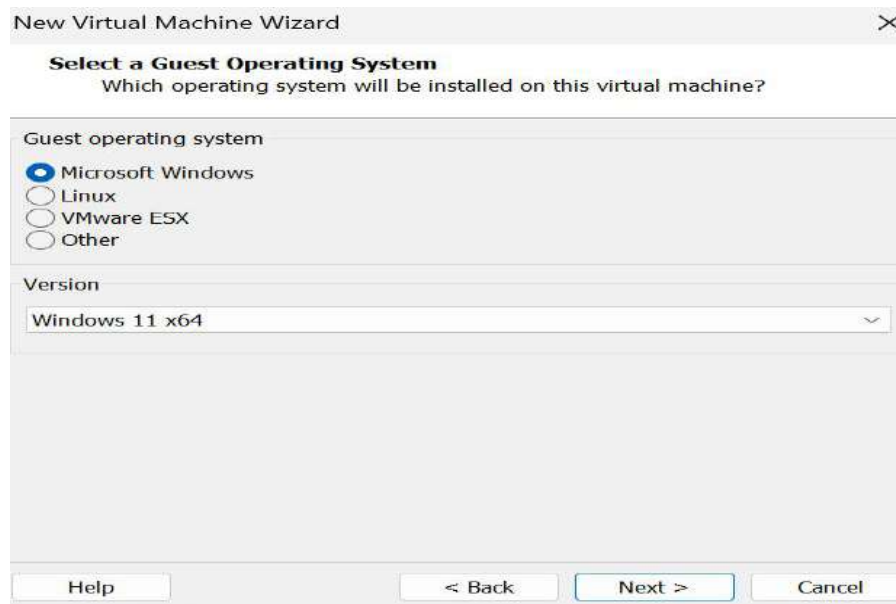


Task 4

Create the Ubuntu Server Virtual Machine

Description:

Set up a new Virtual Machine in VMware Workstation Pro using the Typical configuration setup, mounted the downloaded Ubuntu Server ISO, allocated 20 GB of disk space, and started the VM installation process.



New Virtual Machine Wizard

Easy Install Information
This is used to install Ubuntu 64-bit.

Personalize Linux

Full name:

User name:

Password:

Confirm:

Help < Back Next > Cancel

```

Starting netplan-configure.service - Netplan Backend Configuration...
Starting plymouth-read-write.service - Plymouth To Write Out Runtime Data...
Starting snapd.apparmor.service - ...iles managed internally by snapd...
Starting systemd-binfmt.service - Set Up Additional Binary Formats...
Starting systemd-tmpfiles-setup.service - Create System Files and Directories...
Starting ufw.service - Uncomplicated firewall...
[ OK ] Finished finalrd.service - Create ...untime dir for shutdown pivot root.
[ OK ] Finished plymouth-read-write.service - Plymouth To Write Out Runtime Data.
[ OK ] Finished ufw.service - Uncomplicated firewall.
[ OK ] Finished systemd-tmpfiles-setup.service - Create System Files and Directories.
Starting ldconfig.service - Rebuild Dynamic Linker Cache...
Starting systemd-journal-catalog-update.service - Rebuild Journal Catalog...
[ OK ] Started vgauth.service - Authentic... virtual machines hosted on VMware.
[ OK ] Started open-vm-tools.service - Se... virtual machines hosted on VMware.
[ OK ] Finished systemd-udev-trigger.service - Coldplug All udev Devices.
[ OK ] Started systemd-ask-password-console.service - Requests to Console Directory Watch.
[ OK ] Reached target cryptsetup.target - Local Encrypted Volumes.
[ OK ] Finished systemd-journal-catalog-update.service - Rebuild Journal Catalog.
[ OK ] Finished netplan-configure.service - Netplan Backend Configuration.
[ OK ] Finished kdump-tools.service - Kernel crash dump capture service.
Mounting proc-sys-fs-binfmt_misc.mount - Executable File Formats File System...
[ OK ] Mounted proc-sys-fs-binfmt_misc.mount - Executable File Formats File System.
[ OK ] Finished systemd-binfmt.service - Set Up Additional Binary Formats.
[ 22.310223] piix4_smbus 0000:00:07.3: SMBus Host Controller not enabled!

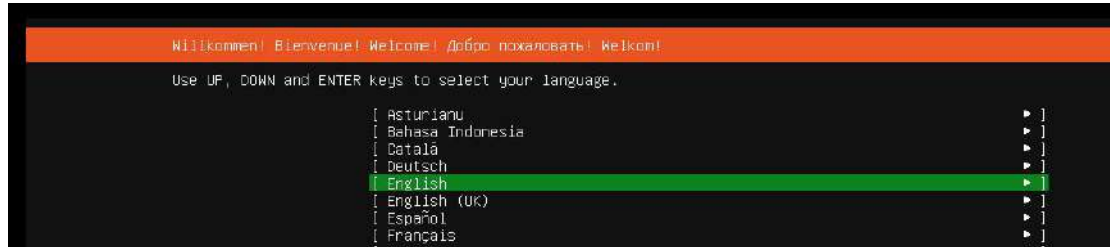
```

Task 5

Choose the Installation Language

Description:

Picked English as the default installation language in the text-based Ubuntu Server setup wizard to continue configuring the OS.



Task 6

Configure the Keyboard Layout

Description:

Selected the base keyboard layout and matching input variant to ensure correct key mapping during system usage.



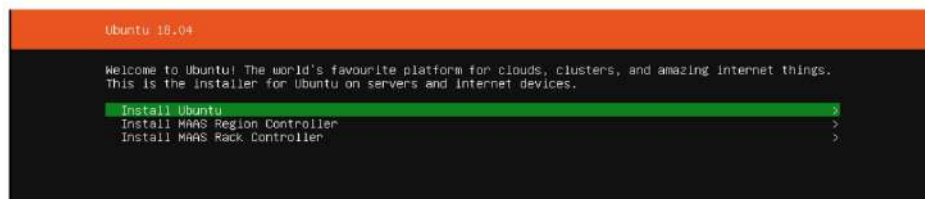
Task 7

Choose the Ubuntu Server Installation Type

Description:

Opted for the standard Ubuntu Server base installation, skipping extra provisioning features like MAAS or automated enterprise tools.

Instructional reference

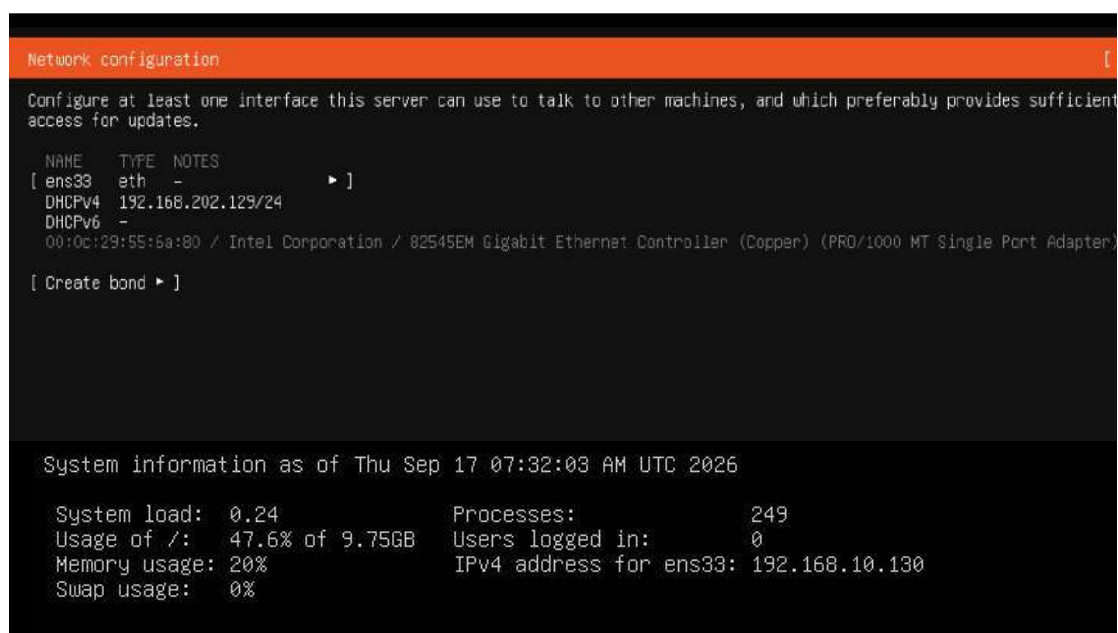


Task 8

Configure the Network

Description:

Ensured the installer detected the network adapter and confirmed it automatically received an IPv4 address via DHCP.



Task 9

Configure Storage

Description:

Selected the guided storage option to automatically partition the VMware virtual disk, then reviewed the resulting storage layout

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY
MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /             10.000G new ext4 new LVM logical volume ▶ ]
[ /boot         1.771G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES
DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 10.222G ▶ ]
free space      8.222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]

USED DEVICES
DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 10.222G ▶ ]
ubuntu-lv       new, to be formatted as ext4, mounted at / 10.000G ▶ ]
[ /dev/sda      local disk 20.000G ▶ ]
partition 1     new, BIOS grub spacer 1.000M ▶ ]
partition 2     new, to be formatted as ext4, mounted at /boot 1.771G ▶ ]
partition 3     new, PV of LVM volume group ubuntu-vg 10.222G ▶ ]

[ Done ]
[ Reset ]
[ Back ]
```

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY
MOUNT POINT    SIZE    TYPE    DEVICE TYPE
[ /             10.000G new ext4 new LVM logical volume ▶ ]
[ /boot         1.771G new ext4 new partition of local disk ▶ ]

AVAILABLE DEVICES
DEVICE          TYPE          SIZE
[ ubuntu-vg (new) LVM volume group 10.222G ▶ ]
free space      8.222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVM) ▶ ]
```

Task 10

Select the Installation Disk

Description:

Chose the target VMware virtual drive for installation and verified its total capacity and details before proceeding with formatting.

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY

  MOUNT POINT    SIZE    TYPE    DEVICE TYPE
  [ /            10,000G new ext4 new LVM logical volume ▶ ]
  [ /boot        1,771G new ext4  new partition of local disk ▶ ]

AVAILABLE DEVICES

  DEVICE                TYPE                SIZE
  [ ubuntu-vg (new)      LVM volume group    18,222G ▶ ]
  free space              8,222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVG) ▶ ]

USED DEVICES

  DEVICE                TYPE                SIZE
  [ ubuntu-vg (new)      LVM volume group    18,222G ▶ ]
  ubuntu-lv             new, to be formatted as ext4, mounted at / 10,000G ▶ ]

[ /dev/sda              local disk                20,000G ▶ ]
partition 1 new, BIOS grub spacer                1,000M ▶ ]
partition 2 new, to be formatted as ext4, mounted at /boot 1,771G ▶ ]
partition 3 new, PV of LVM volume group ubuntu-vg 18,225G ▶ ]

[ Done ]
[ Reset ]
[ Back ]
```

```
Storage configuration [ Help ]

FILE SYSTEM SUMMARY

  MOUNT POINT    SIZE    TYPE    DEVICE TYPE
  [ /            10,000G new ext4 new LVM logical volume ▶ ]
  [ /boot        1,771G new ext4  new partition of local disk ▶ ]

AVAILABLE DEVICES

  DEVICE                TYPE                SIZE
  [ ubuntu-vg (new)      LVM volume group    18,222G ▶ ]
  free space              8,222G ▶ ]

[ Create software RAID (md) ▶ ]
[ Create volume group (LVG) ▶ ]

USED DEVICES

  DEVICE                TYPE                SIZE
  [ ubuntu-vg (new)      LVM volume group    18,222G ▶ ]
  ubuntu-lv             new, to be formatted as ext4, mounted at / 10,000G ▶ ]

[ /dev/sda              local disk                20,000G ▶ ]
partition 1 new, BIOS grub spacer                1,000M ▶ ]
partition 2 new, to be formatted as ext4, mounted at /boot 1,771G ▶ ]
partition 3 new, PV of LVM volume group ubuntu-vg 18,225G ▶ ]

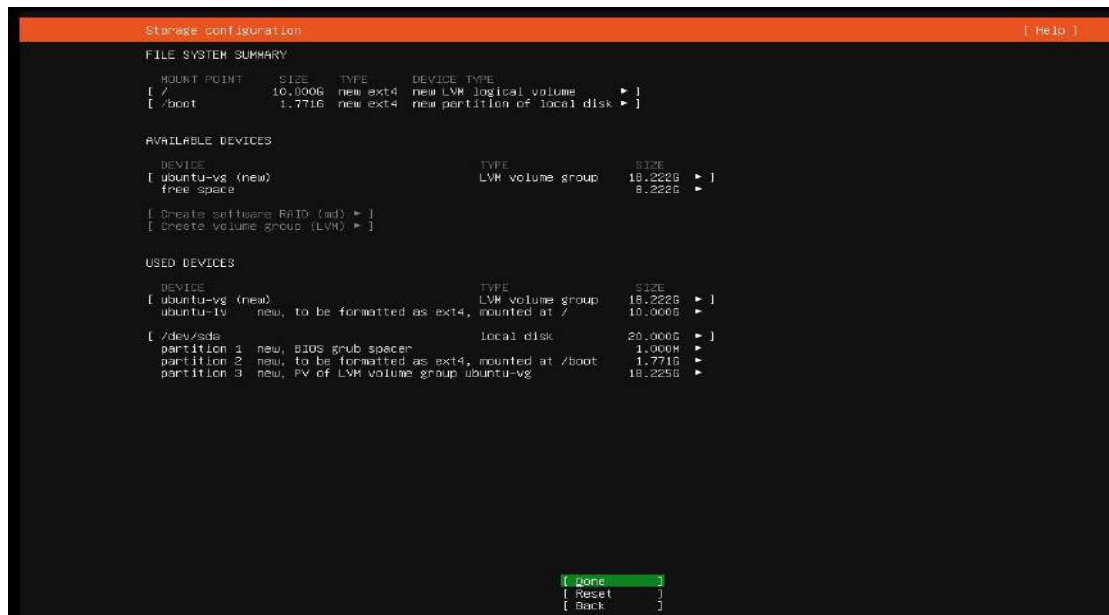
[ Done ]
[ Reset ]
[ Back ]
```

Task 11

Review the Partitions

Description:

Examined the generated file system layout and mount points created by the installer to confirm proper drive partitioning on the virtual machine

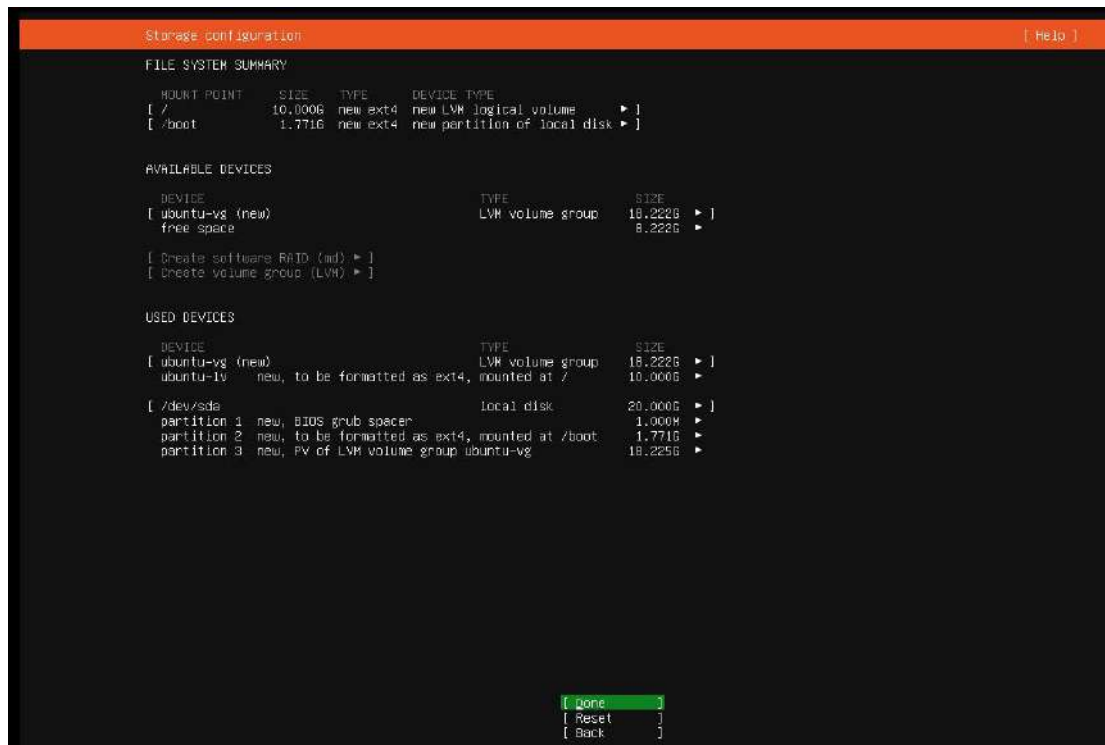


Task 12

Confirm the Storage Changes

Description:

Accepted the warning prompt regarding drive formatting and confirmed the partition scheme to begin writing changes to the virtual disk.



Task 13

Configure the Ubuntu Profile

Description:

Entered the primary administrator credentials and server hostname in the installation wizard, matching the actual student identity and assigned GitHub handle.

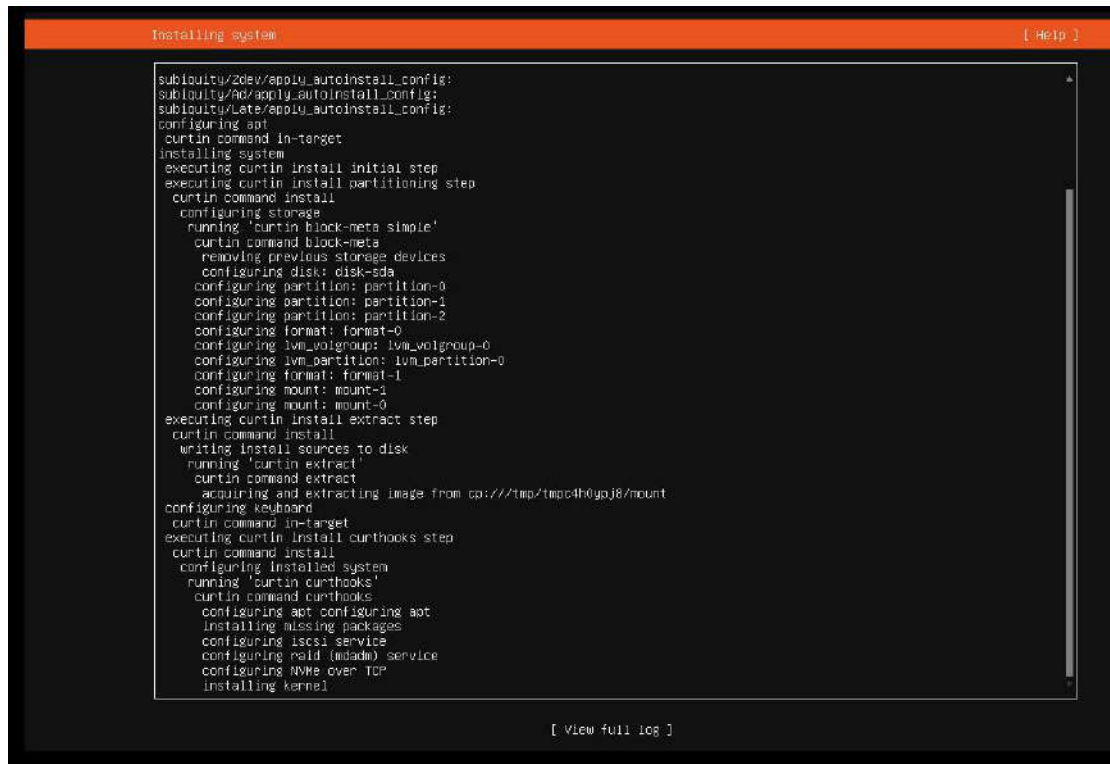


Task 14

Complete the Software Installation

Description:

Started the base OS package extraction and setup process, letting it run without interruption until the core system installation was finished.



```
Installing system [ Help ]
subiquity/2dev/apply_autoinstall_config:
subiquity/80/apply_autoinstall_config:
subiquity/late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
curtin command install
  configuring storage
    running 'curtin block-meta simple'
    curtin command block-meta
      removing previous storage devices
      configuring disk: disk-sda
      configuring partitions: partition-0
      configuring partitions: partition-1
      configuring partitions: partition-2
      configuring format: format-0
      configuring lvm_voigroup: lvm_voigroup-0
      configuring lvm_partition: lvm_partition-0
      configuring format: format-1
      configuring mount: mount-1
      configuring mount: mount-0
  executing curtin install extract step
  curtin command install
    writing install sources to disk
    running 'curtin extract'
    curtin command extract
      acquiring and extracting image from cp:///tmp/tmpc4h0yaJ8/mount
  configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring naid (mdadm) service
    configuring NVMe over TCP
    installing kernel

[ View full log ]
```

Task 15

Reboot and Verify the Ubuntu Identity

Description:

Restarted the virtual machine after installation and logged into the command-line terminal to verify the custom server hostname and shell prompt settings.

[Installation complete]

[Help]

```
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
  acquiring and extracting image from cp:///tmp/tmpc4h0ypja/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
  configuring apt
  configuring apt
  installing missing packages
  configuring iscsi service
  configuring raid (mdadm) service
  configuring NVMe over TCP
  installing kernel
  setting up swap
  apply networking config
  writing etc/istab
  configuring multipath
  updating packages on target system
  configuring pollinate user-agent on target
  configuring kernel crash dumps settings
  final kernel configuration
  configuring target system bootloader
  installing grub to target devices
  copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:
```

[View full log]

[Reboot Now]

```
[ 24.770599] cloud-init[1171]: The key's randomart image is:
[ 24.770662] cloud-init[1171]: +---[ECDSA 256]---+
[ 24.770690] cloud-init[1171]: |                    |
[ 24.770724] cloud-init[1171]: |                    |
[ 24.770748] cloud-init[1171]: |                    |
[ 24.770770] cloud-init[1171]: |                    |
[ 24.773264] cloud-init[1171]: | S oo+ B.= |
[ 24.773681] cloud-init[1171]: | .+o+.@o |
[ 24.773946] cloud-init[1171]: | = o*== |
[ 24.774125] cloud-init[1171]: | +.*oX+. |
[ 24.774259] cloud-init[1171]: | .+O+B++E |
[ 24.774442] cloud-init[1171]: +----[SHA256]-----+
[ 24.774697] cloud-init[1171]: Generating public/private ed25519 key pair.
[ 24.777953] cloud-init[1171]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 24.780098] cloud-init[1171]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 24.782195] cloud-init[1171]: The key fingerprint is:
[ 24.784347] cloud-init[1171]: SHA256:/24Mh1x27xMI74IWqHpYKCdtSnqkXUphmrPw7vC7nfk root@ubuntu
[ 24.784988] cloud-init[1171]: The key's randomart image is:
[ 24.787380] cloud-init[1171]: +---[ED25519 256]---+
[ 24.790242] cloud-init[1171]: |                    |
[ 24.793243] cloud-init[1171]: |                    |
[ 24.793861] cloud-init[1171]: |                    |
[ 24.794362] cloud-init[1171]: | o . . o o o |
[ 24.794662] cloud-init[1171]: | oo.. . .S= . . . |
[ 24.794847] cloud-init[1171]: | B o B o =.+ . o |
[ 24.795033] cloud-init[1171]: | */ + . .+. . |
[ 24.795171] cloud-init[1171]: | B o =..o .o |
[ 24.795297] cloud-init[1171]: | +*+..E oo |
[ 24.795424] cloud-init[1171]: +----[SHA256]-----+

ubuntu login: ^^_
```

```
ubuntu login: laibasaleem066
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 04:55:09 AM UTC 2026

System load:  0.0           Processes:      221
Usage of /:   50.7% of 9.75GB Users logged in: 0
Memory usage: 22%          IPv4 address for ens33: 192.168.10.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

laibasaleem066@ubuntu:~$ _
```

```
ubuntu login: laibasaleem066
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 04:55:09 AM UTC 2026

System load:  0.0           Processes:      221
Usage of /:   50.7% of 9.75GB Users logged in: 0
Memory usage: 22%          IPv4 address for ens33: 192.168.10.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
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Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

laibasaleem066@ubuntu:~$ _
```

Task 16

Find the Ubuntu Server IP Address

Description:

Ran network diagnostic commands in the terminal to discover the dynamically allocated IPv4 address on the active virtual network adapter.

```
laibasaleem066@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
```

```
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:fb:ab:7f brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29fbab7f
    inet 192.168.10.130/24 metric 100 brd 192.168.10.255 scope global dynamic ens33
        valid_lft 1596sec preferred_lft 1596sec
    inet6 fe80::28c:29ff:fe7b:ab7f/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
laibasaleem066@ubuntu:~$
```

Task 17

Connect to Ubuntu Server from Windows

Description:

Initiated an SSH session from the Windows host using PowerShell/Command Prompt to connect to the VM's IP, accepted the security host key fingerprint, and successfully logged in.

```
C:\Users\User>ssh laibasaleem066@192.168.10.130
```

```
C:\Users\User>ssh laibasaleem066@192.168.10.130
The authenticity of host '192.168.10.130 (192.168.10.130)' can't be established.
ED25519 key fingerprint is SHA256:/24MhlxZ7xMI74IWqHpYKCdtSnqkWUphmrPw7vC7nfk.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

```
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 06:32:51 AM UTC 2026

System load:  0.08               Processes:            223
Usage of /:   47.6% of 9.75GB    Users logged in:     0
Memory usage: 25%               IPv4 address for ens33: 192.168.10.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

laibasaleem066@ubuntu:~$ |
```

Task 18

Prepare Lab1 Solution PDF

Description:

Assembled all completed task summaries, image captions, and verification screenshots into a clean, formatted report featuring an official cover page, then saved the final document as a PDF in the designated submission directory.

Lab Number & Title: Lab 01 - Ubuntu Server Installation

Student Name: Laiba Saleem

Registration Number: 2024-BSE-66

Course Code: SE-665

Section: Section B

GitHub Username: LaibaSaleem066



Lab No 1

Cloud Computing

**Lab Title: GitHub Identity, Student Portal Registration,
and Ubuntu Server Installation**

Submitted To: Engr. Waqas Saleem

Submitted By: Laiba Saleem

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GitHub Username: LaibaSaleem066
